OK FOOTWEAR LIMITED

Enterprise Resource Planning System

Technical System Design Document

Version 1.0 | May 2025

Document Title	OK Footwear ERP — Technical System Design
Version	1.0
Status	Draft — For Management Review
Prepared By	ERP Design Team
Company	OK Footwear Limited
Date	May 2025
Classification	Confidential — Internal Use Only

Table of Contents

1. Executive Summary

OK Footwear Limited is a Bangladesh-based private limited company engaged in the manufacturing and export of footwear. The company operates across procurement, production, outsourcing, and delivery, with obligations under the Bangladesh Companies Act 1994, Labour Act 2006, and NBR tax regulations.

This document presents the complete technical design of a bespoke Enterprise Resource Planning (ERP) system purpose-built for OK Footwear. The system replaces fragmented spreadsheets and manual processes with an integrated, web-based platform covering the full business lifecycle — from buyer order through production, quality control, shipment, HR and payroll, finance, and board governance.

Key Design Highlights

- Modular Monolith architecture — a single deployable NestJS application with clean, enforced module boundaries. Chosen for operational simplicity at 50–200 concurrent users, full ACID transaction support across modules, and a clear path to microservice extraction if scale demands it.
- React 18 frontend — bilingual (English and Bangla), responsive, and role-gated. Three client surfaces: full web browser ERP, factory tablet touch interface, and a mobile employee self-service portal.
- PostgreSQL 16 database — 8 schemas, 67 business tables, financial-grade ACID compliance, double-entry GL with period locking, and trigger-maintained stock balances.
- Full QA automation — no manual QA. Unit, integration, database, contract, E2E (Playwright), performance (k6), and security tests run in a fully automated GitHub Actions CI/CD pipeline.
- Bangladesh legal compliance built in — Labour Act 2006 gratuity formula, NBR TDS/WHT deductions, Companies Act 1994 AGM rules, RJSC director registration.

2. Company & Project Background

2.1 Company Profile

Company Name	OK Footwear Limited
Entity Type	Private Limited Company — Bangladesh Companies Act 1994
Business	Footwear manufacturing and export
Buyers	Bay, Lotto, Walker and others (domestic & export)
Operations	Procurement, in-house production, third-party outsourcing, import of machinery
Regulatory	RJSC, NBR, Bangladesh Bank, Labour Directorate, BGMEA
Staff	Factory operators, office staff, management, board directors

2.2 Business Problem

OK Footwear currently manages operations across disconnected spreadsheets, paper forms, and standalone tools. This creates critical gaps:

- No real-time inventory visibility — stock levels unknown until physical count
- Payroll computed manually in Excel — error-prone, no PF/TDS audit trail
- Order-to-delivery tracking done on paper — milestone delays undetected
- Board governance (AGM, resolutions, shares) maintained in unsecured files
- No RBAC — all staff see all data regardless of role or department
- Zero integration between finance, HR, procurement, and production

2.3 Project Objectives

- Digitalise and integrate all 14 functional areas into a single system
- Enforce role-based access control across all modules and data
- Ensure full compliance with Bangladesh Companies Act, Labour Act, and NBR requirements
- Eliminate manual payroll computation and automate GL posting for all financial transactions
- Provide real-time stock visibility with automated reorder alerts
- Deliver a mobile ESS portal so factory workers can apply leave and download payslips
- Achieve full QA automation with zero manual testing

3. System Objectives & Scope

3.1 Modules In Scope

#	Module	Key Capabilities
1	Order Management	Buyer master, article catalogue, bulk/sample orders, size breakdown, milestones, quotations, complaints & CAPA
2	Procurement	Vendor qualification, PRs, multi-level PO approval, GRN three-way match, vendor invoices
3	BOM & Costing	Versioned BOM, cost sheets, wastage %, variance tracking, approved BOM gate for production
4	Manufacturing	Factory lines, production orders, daily output, QC (AQL), efficiency tracking, scrap management
5	Outsourcing	Subcontractor work orders, material issue/return, capacity overflow management
6	Inventory	Multi-warehouse, append-only ledger, trigger-maintained balances, stock counts
7	Delivery & Dispatch	Delivery challans, proof of delivery, buyer returns
8	Import/Export	Import & export LC lifecycle, landed cost, duty/customs tracking
9	Finance	Double-entry GL, AP/AR, fixed assets with depreciation, budgets, bank reconciliation
10	HR & Payroll	Employee lifecycle, leave, attendance, payroll (LOP/OT/PF/TDS), PF, gratuity, ESS/MSS portal
11	Board Governance	Director register, share register, board meetings, AGM, resolutions (SHA-256 hash), dividends (WHT)
12	System Admin	Users, roles, RBAC, audit log, compliance register, 2FA, notifications

3.2 Out of Scope (Initial Release)

- Customer-facing e-commerce or buyer portal
- Machine learning or AI-based demand forecasting
- Real-time bank API integration (file-based ACH export used instead)
- Native mobile applications (React PWA used for mobile access)

4. Architecture Overview

4.1 Architecture Decision — Modular Monolith

After evaluating Microservices, Modular Monolith, and Traditional Monolith architectures, the team selected a Modular Monolith for the following reasons:

Factor	Microservices	Modular Monolith ✓	Traditional Monolith
50–200 concurrent users	✗ Over-engineered	✓ Right-sized	✓ Works but messy
Cross-module ACID transactions	✗ Saga complexity	✓ Simple transactions	✓ Works but coupled
Team size (3–6 devs)	✗ High DevOps overhead	✓ Manageable	✓ Easy initially
Payroll (multi-module)	✗ Very complex	✓ One transaction	✓ Works but messy
Operational complexity	✗ K8s, mesh, tracing	✓ One container	✓ Simple
Future scalability	✓ Best path	✓ Extractable modules	✗ Hard to split
Verdict	✗ Wrong tool	✓ SELECTED	✗ Technical debt

4.2 System Layers

Layer	Components	Technology
Client	Web browser, Factory tablet, Mobile ESS portal	React 18 PWA
Edge	SSL/TLS, reverse proxy, rate limiting, static serving	Nginx
Frontend App	SPA, routing, components, state, API client	React 18 + TypeScript + Vite
Backend App	HTTP pipeline, 8 NestJS modules, shared kernel, jobs	NestJS + TypeScript
Data Store	Primary RDBMS, queues + cache, document/file storage	PostgreSQL 16 + Redis 7 + S3
External	Salary disbursement, SMS, email, biometric, eSign	Bank / SSL Wireless / SES / DocuSign
Infra	Containers, CI/CD, error tracking, observability	Docker + K8s + GitHub Actions + Grafana

4.3 Architecture Diagram

The interactive system architecture diagram is provided as a separate HTML file (OK_Footwear_ERP_System_Architecture.html). The diagram shows all 7 layers, the frontend and backend columns communicating through a REST API boundary, data stores, external services, and infrastructure.

Key data flows:

- Client → Nginx: All traffic over HTTPS / TLS 1.3
- Nginx → React: Serves pre-built static files (SPA mode for all non-/api routes)
- Nginx → NestJS: Reverse proxies /api/* to port 3000
- React ↔ NestJS: REST/JSON with JWT Bearer auth; SSE stream for real-time notifications
- NestJS → PostgreSQL: Type-safe queries via Prisma ORM through PgBouncer pool
- NestJS → Redis: BullMQ queues (DB0), auth cache (DB1), app cache (DB2)

- NestJS → S3/MinIO: PDFs and Excel files streamed directly to object storage
- NestJS → External: Bank ACH files, SMS OTP, email, biometric sync, DocuSign eSign

5. Technology Stack

5.1 Frontend

Technology	Version	Purpose
React	18	UI framework — component-based SPA
TypeScript	5.x	Type safety — mandatory for ERP-scale codebase
Vite	5.x	Build tool — fastest HMR, code splitting per route
Tailwind CSS	3.x	Styling — utility-first, consistent design system
shadcn/ui	Latest	Component library — Radix primitives + Tailwind
TanStack Query	v5	Server state — caching, background refetch, optimistic updates
TanStack Table	v8	Data grids — virtual scrolling for large datasets
Zustand	4.x	Client state — auth, UI preferences, notifications
React Hook Form	v7	Form management — performance + field-level validation
Zod	3.x	Schema validation — runtime + compile-time type safety
Recharts	2.x	Charts — KPI dashboards and production analytics
i18next	Latest	Internationalisation — English/Bangla bilingual support
Axios	1.x	HTTP client — JWT interceptors and auto-refresh
Playwright	Latest	E2E testing — cross-browser full automation
Vitest + RTL	Latest	Unit testing — Vite-native, React Testing Library

5.2 Backend

Technology	Version	Purpose
NestJS	10.x	Backend framework — module-based, TypeScript-first
TypeScript	5.x	Type safety — end-to-end with frontend
Express	4.x	HTTP adapter — larger middleware ecosystem vs Fastify
Prisma ORM	v5	Database ORM — multi-schema, type-safe, managed migrations
PgBouncer	Latest	Connection pooling — transaction mode, 20 connection pool
BullMQ	5.x	Job queues — payroll, PDF, email, SMS, reports
@nestjs/schedule	Latest	Cron jobs — daily/monthly with Redis distributed lock
Puppeteer	Latest	PDF generation — headless Chrome, pixel-perfect output
ExcelJS	4.x	Excel exports — .xlsx reports streamed to S3
Argon2	Latest	Password hashing — memory-hard, more secure than bcrypt
otplib	Latest	TOTP 2FA — Google Authenticator compatible
nestjs-pino	Latest	Structured JSON logging — correlation-id, Grafana Loki

Technology	Version	Purpose
Jest + Supertest	Latest	Testing — unit + integration with real DB in CI

5.3 Data Stores

Store	Version	Usage	Key Configuration
PostgreSQL	16	Primary RDBMS	8 schemas, 67 tables, PgBouncer pool, WAL to S3, read replica for reports
Redis	7	Queues, cache, sessions	DB0 queues, DB1 auth, DB2 app cache; RDB + AOF persistence; allkeys-lru
S3 / MinIO	-	Object/file storage	3 buckets: documents (7yr), exports (30d), receipts (90d); SSE-S3 encryption

6. Frontend Architecture

6.1 Client Surfaces

Surface	Target Users	Access Method	Key Routes
Full ERP	Office staff, managers, finance, HR, board	Desktop/laptop browser	/dashboard, /orders, /finance, /hr, /board, /system
Factory Tablet	Production supervisors, QC inspectors	10" Android/iOS tablet	/manufacturing/daily-entry, /manufacturing/qc-form
Mobile ESS	All active employees	iOS/Android browser	/ess/leave, /ess/payslip, /ess/expense, /ess/attendance

6.2 Frontend Layers

Layer	Technology	Responsibility
Build	Vite 5 + TypeScript	Bundling, HMR, code splitting per route, tree-shaking
Routing	React Router v6	Lazy-loaded, role-gated routes via RoleGuard HOC
Pages	React components	4 groups: Operations, Management, ESS, MSS — all role-gated
Components	shadcn/ui + Tailwind	Base UI primitives; TanStack Table; custom ERP widgets
Forms	React Hook Form + Zod	Type-safe forms; server RFC 7807 errors auto-mapped to fields
Server State	TanStack Query v5	Caching, background refetch, optimistic updates, invalidation
Client State	Zustand	authStore, uiStore (theme/locale), notifStore (SSE badges)
API Client	Axios	JWT auto-attach; silent 401 refresh; typed request/response DTOs
Real-time	SSE EventSource	/api/notifications/stream for in-app push alerts
i18n	i18next	English/Bangla JSON bundles; lazy-loaded; locale in localStorage

7. Backend Architecture

7.1 HTTP Request Pipeline

Every incoming HTTP request traverses seven middleware layers before reaching the controller:

Step	Layer	Responsibility
1	JwtAuthGuard	Validates Bearer token; decodes user identity and role list
2	RbacGuard	Checks module:action permission against Redis-cached permissions (5-min TTL)
3	ThrottlerGuard	Enforces Redis sliding-window rate limits (100 req/min per user)
4	ValidationPipe	Validates class-validator DTOs; strips unknown properties (whitelist: true)
5	Controller	Routes request to the appropriate service method
6	ResponseInterceptor	Wraps success response in { data, timestamp } standard envelope
7	HttpExceptionFilter	Catches all errors; formats RFC 7807 problem+json with correlation_id

7.2 Shared Kernel

Service	Responsibility
AuthService	JWT generation/validation, Argon2id hashing, TOTP 2FA, RBAC permission cache in Redis
Event Bus	@nestjs/event-emitter — in-process synchronous events between modules (zero network hop)
Logger (Pino)	nestjs-pino structured JSON logging; correlation-id per request; shipped to Grafana Loki
ConfigService	@nestjs/config with Joi validation; all env vars validated at startup; namespaced configs

7.3 Background Processing

Queue / Job	Timeout	Triggered By	Output
payroll-compute	30 min	HR Manager payroll run	Employee entries + GL journal + bank disbursement file
pdf-generation	60 sec	Payslip, challan, minutes	PDF streamed to S3; pre-signed URL returned to client
email-dispatch	30 sec	Any module notification	SES email with 3 retries and exponential backoff
sms-dispatch	15 sec	2FA OTP, payroll alerts	SSL Wireless SMS delivered to Bangladesh number
report-export	5 min	Scheduled report requests	ExcelJS .xlsx streamed to S3; pre-signed URL returned
daily-cron	10 min	Nightly 02:00 BDT	Depreciation post, stock summary refresh, compliance check
monthly-cron	20 min	1st of month 03:00 BDT	Gratuity accrual, leave carry-forward, PF interest credit

8. Database Design

8.1 Schema Organisation

Schema	Tables	Domain	Key Tables
sys	8	System & Admin	users, roles, permissions, audit_logs, notifications, compliance_items
ord	9	Order Management	buyers, articles, orders, order_lines, order_milestones, quotations, complaints
prc	10	Procurement	vendors, purchase_orders, po_lines, goods_receipts, gr_lines, vendor_invoices
mfg	15	Manufacturing	bom_headers, bom_lines, production_orders, daily_productions, qc_results, machines
inv	8	Inventory	stock_items, warehouses, stock_transactions, stock_balances, stock_counts
fin	19	Finance	chart_of_accounts, gl_entries, gl_entry_lines, fixed_assets, budgets, import_lcs
hr	26	HR & Payroll	employees, payroll_runs, payroll_entries, leave_requests, attendance_records
brd	13	Board Governance	directors, shareholders, share_transactions, board_meetings, resolutions, dividends

8.2 Key Design Patterns

Pattern	Applied To	Purpose
UUID primary keys	All 67 tables	No sequential IDs leaked; safe for client generation; future sharding
Soft delete (deleted_at)	All master data tables	History preservation; performance via partial index WHERE deleted_at IS NULL
Append-only ledgers	stock_transactions, gl_entry_lines, pf_transactions	Immutable financial records; no UPDATE/DELETE ever allowed
Status machine + CHECK	Orders, POs, payroll runs, meetings	Invalid transitions caught at DB level; not just application level
Encrypted sensitive fields	NID, passport, bank accounts	AES-256-GCM at app layer; stored as BYTEA in *_secrets tables
Trigger-maintained balances	inv.stock_balances	Running stock balance updated on every transaction insert; O(1) read
Date-range partitioning	6 high-volume tables	Query performance; audit_logs, stock_transactions, gl_entry_lines, attendance
Materialized views	shareholding, stock_summary	Pre-aggregated for fast dashboard queries; refreshed nightly by cron
GIN trigram indexes	buyer, vendor, employee names	Fuzzy search for type-ahead dropdowns without Elasticsearch

8.3 Bangladesh Compliance in Schema

Compliance Requirement	Database Implementation
Labour Act 2006 — Gratuity	hr.compute_gratuity() function: Basic × (30/26) × years; ≥6 months rounds up
Companies Act 1994 — AGM 15-month rule	agms table with last_agm_date; application enforces 15-month interval
Companies Act — Register of Members	brd.shareholders permanent table; share_transactions append-only ledger
Companies Act — Register of Directors	brd.directors permanent table; NID/passport AES-256-GCM encrypted
NBR TDS on vendor payments	prc.vendor_invoices.tds_amount; auto-deducted before net_payable
NBR WHT on dividends	brd.dividends.withholding_tax_pct; brd.dividend_payments.tax_deducted auto-computed
Board resolutions tamper-evidence	brd.resolutions.sha256_hash stored on Chairman eSign; column never mutable

9. Module Descriptions

9.1 Order Management

Sub-module	Description
Buyers	Master record for all buying companies with credit limits, payment terms, currency
Articles	Shoe style catalogue with category, size system, season metadata
Orders	Status machine: draft→confirmed→in_production→qc→packed→delivered. Sample gate + BOM gate
Quotations	Buyer price proposals with auto-populated cost sheet. Win/loss tracking
Samples	Multi-round tracking (PP, counter, size set, TOP). Approval gate blocks bulk production
Milestones	Auto-generated backward schedule from delivery date: material booking, start, QC, packing, ship
Complaints	Post-delivery quality/shortage complaints with CAPA workflow and action tracking

9.2 Procurement

Sub-module	Description
Vendors	Approved vendor master with status, performance rating, TIN for TDS calculation
Purchase Orders	Multi-level approval by amount threshold. Only approved vendors receive POs
GRN	Goods Receipt Note with QC inspection. Triggers inventory stock insert on approval
Vendor Invoices	Three-way match: PO ≥ GRN ≥ invoice. TDS deducted. Links to GL AP posting

9.3 Manufacturing

Sub-module	Description
BOM	Versioned Bill of Materials. Approval workflow. Wastage % per component. Size-specific quantities
Cost Sheets	Auto-generated from BOM + vendor rates. Material/labour/overhead split. Variance vs actual
Production Orders	Links sales order to factory line with BOM version. Planned vs produced tracking
Daily Production	Per-operation, per-shift output entry. Efficiency % generated column. Nightly lock
QC Results	Inline and final QC with AQL sampling. Defect details in JSONB. Pass/fail/rework verdict
Machines	Machine register with maintenance log. Downtime hours. Links to fixed asset for depreciation
Scrap	Scrap recording by type. Disposal authorisation workflow. Sale/recycle/landfill tracking

9.4 Inventory

Sub-module	Description
Stock Items	Item master with category, UOM, reorder level, lead time. GIN trigram search
Warehouses	Physical locations by type: raw material, finished goods, packing, accessories

Sub-module	Description
Stock Transactions	Append-only ledger. Direction +1/-1. Weighted average cost recalculated on receipt
Stock Balances	PostgreSQL trigger-maintained after every transaction insert. CHECK prevents negative
Stock Counts	Physical count with system snapshot. Variance workflow with Finance Manager approval

9.5 Finance

Sub-module	Description
General Ledger	Double-entry journal with period locking. Balance validation trigger. Central posting API
Fixed Assets	Asset register with monthly SL/DB depreciation cron auto-GL post. NBV generated column
Budgets	Annual OPEX/CAPEX budgets. Monthly line items. Budget vs actual variance from GL
Bank Accounts	Company bank register. Statement import for reconciliation. Links to GL cash accounts
Import LCs	Full LC lifecycle from opening to settlement. Landed cost computation per shipment
Export LCs	Export LC from receipt to repatriation. Tolerance %, document submission, payment tracking
Buyer Invoices	AR invoices with ageing. Collection tracking. Dispute workflow via complaint module

9.6 HR & Payroll

Sub-module	Description
Employees	Full lifecycle: hire, confirm, transfer, promote, exit. Sensitive fields AES-256 encrypted
Payroll	LOP/OT calculation, PF, TDS per NBR, festival bonus. GL post on disbursement. Bank file export
Leave	Per-type policies (accrual, carry-forward, encashment). Balance ledger. Conflict detection
Attendance	Biometric device sync (webhook or polling cron). Manual correction. LOP calculation
PF	Per-employee PF account. Monthly contributions. Annual interest credit cron. Statement
Gratuity	Monthly provision via hr.compute_gratuity() function. Labour Act 2006 formula
ESS Portal	Employee self-service: leave apply, payslip PDF download, expense claim, attendance view
MSS Portal	Manager self-service: approval queue with SLA badges, team leave calendar

9.7 Board Governance

Sub-module	Description
Directors	RJSC Register of Directors. DIN storage. NID/passport AES-256 encrypted
Shareholders	Register of Members per Companies Act. Share certificate issuance and cancellation
Share Register	Append-only share transaction ledger (allotment, transfer, buyback). Materialized % view
Board Meetings	Scheduling with quorum validation. Minutes blocked if inquorate. eSign integration
Resolutions	Register with SHA-256 tamper-evident hash on signed document. Ordinary/Special/Circular
AGM	15-month interval rule enforced. Notice dispatch tracking. Proxy registration
Dividends	Interim/final declaration. Auto-computation per shareholder with WHT per NBR rules

9.8 System Administration

Sub-module	Description
Authentication	JWT + Argon2id + TOTP 2FA. httpOnly refresh cookies. Account lockout after 5 failures
RBAC	12 seeded roles → permissions (module:action). Redis-cached 5 min. Invalidated on change
Audit Log	Append-only sys.audit_logs for all write operations with old/new JSONB values. GIN search
Notifications	In-app (SSE), email, SMS channels. BullMQ async dispatch. Unread badge count
Compliance	Licence/certificate expiry register. Configurable alert days. Nightly check cron

10. Security Design

10.1 Authentication

Aspect	Implementation
Password hashing	Argon2id — memory-hard, more secure than bcrypt for modern hardware
Access tokens	JWT HS256; 8-hour expiry; stored in browser memory (not localStorage)
Refresh tokens	30-day JWT in httpOnly + Secure + SameSite=Strict cookie; rotated on each use
2FA	TOTP via otplib; Google Authenticator compatible; required for MD and Finance Manager
Account lockout	After 5 consecutive failed logins; locked for 30 minutes; stored in PostgreSQL
Session invalidation	Refresh token blacklisted in Redis (TTL = token remaining life) on logout

10.2 Authorisation

Aspect	Implementation
Model	Role-Based Access Control (RBAC) with fine-grained module:action permissions
Enforcement	RbacGuard on every NestJS controller; permissions cached in Redis (5-min TTL)
Frontend enforcement	Client-side route guards using same RBAC model; backend also enforces independently
IDOR prevention	All repository queries scope data to the authenticated user's entity context
Permission invalidation	Redis permission cache cleared immediately on any role assignment change

10.3 Data Security

Aspect	Implementation
Sensitive field encryption	NID, passport, bank accounts: AES-256-GCM; stored as BYTEA in *_secrets tables
SQL injection prevention	Prisma parameterised queries; ValidationPipe strips unknown fields
Transport security	TLS 1.3 at Nginx; HSTS header; HTTP redirected to HTTPS
File security	S3 presigned URLs (15-min TTL); no direct public bucket access; SSE-S3 encryption
Audit trail	Append-only sys.audit_logs captures all write operations with before/after JSONB

10.4 Infrastructure Security

- Secrets managed via environment variables — never committed to code or Docker images
- GitHub Secrets for CI/CD pipeline credentials — rotated quarterly
- PostgreSQL and Redis not exposed to internet — accessible only from application pods
- S3 bucket policies block all public access; presigned URLs used for all downloads
- Rate limiting at Nginx (per-IP) and application layer (per-user via Redis)
- CORS configured to allow only the production application origin

11. Testing Strategy

The project follows a full QA automation strategy — no manual testing. All test layers run unattended in the CI/CD pipeline. A failing test in any layer blocks the PR from merging.

11.1 Test Pyramid

Layer	Tool	Coverage Target	When
Unit tests (50%)	Jest + ts-jest / Vitest + RTL	≥85% statements	Every commit
Database tests	Jest + real PostgreSQL	100% triggers/functions	Every commit
Integration tests (30%)	Supertest + testcontainers	≥90% endpoints	Every PR
Contract tests (10%)	Pact (consumer + provider)	All API contracts	Every PR
E2E tests (10%)	Playwright (Chromium + Firefox)	All critical flows	Merge to main
Performance tests	k6	p95 < 500ms	Merge to main
Security tests	Jest suites + OWASP ZAP	22 security cases	Nightly

11.2 Coverage Thresholds (Enforced in CI)

Layer	Statements	Branches	Functions
Backend services	≥85%	≥80%	≥85%
Backend controllers	≥90%	≥85%	≥90%
Guards & pipes	≥95%	≥90%	≥95%
DB triggers/functions	100%	100%	100%
Frontend components	≥80%	≥75%	≥80%
Zod schemas	100%	100%	100%
Overall	≥80%	≥75%	≥80%

12. Infrastructure & Deployment

12.1 Environments

Environment	Purpose	Infrastructure
Local development	Developer workstation; full stack in one command	Docker Compose — PostgreSQL + Redis + MinIO + NestJS + Nginx
Staging	Pre-production; E2E, performance, security target	AWS EKS small cluster — mirrors production configuration
Production	Live system for OK Footwear operations	AWS EKS + RDS PostgreSQL 16 + ElastiCache Redis + S3

12.2 CI/CD Pipeline

Stage	Action	Gate
1. Lint + Type-check	ESLint, TypeScript compiler	Fails on any lint or type error
2. Unit tests	Jest + Vitest with coverage enforcement	Fails below coverage threshold
3. Integration tests	Supertest + testcontainers (PG + Redis)	Fails on any test failure
4. Contract tests	Pact consumer + provider verification	Fails on contract mismatch
5. Docker build	Multi-stage build; image size check	Fails on build error
6. Push to ECR	Push image to AWS Elastic Container Registry	Only on merge to main
7. K8s deploy	kubectl rollout + prisma migrate init container	Only on merge to main
8. E2E tests	Playwright against staging environment	Only on merge to main
9. Performance tests	k6 load + spike tests against staging	Only on merge to main

12.3 Observability

Tool	What It Monitors	Alert Condition
Prometheus + Grafana	HTTP latency p50/p95/p99, BullMQ queue depth, DB pool connections, error rate	p95 > 3s or error rate > 1%
Grafana Loki	Structured JSON logs via Promtail sidecar; correlation-id linking across requests	Error log spike
Sentry	Frontend JS errors + backend 5xx exceptions with full context and source maps	New error fingerprint type

13. Implementation Phases & Timeline

Phase	Duration	Modules	Key Deliverables
Phase 1 — Core Operations	Months 1–4	Orders, Procurement, Inventory, Finance (GL core), Delivery	Order creation to dispatch; stock management; basic GL; PO and GRN workflow
Phase 2 — Production & HR	Months 5–8	Manufacturing (BOM, Prod, QC), HR & Payroll, ESS/MSS Portal	Factory production tracking; payroll runs; leave management; employee self-service portal
Phase 3 — Compliance	Months 9–12	Import/Export, Fixed Assets, Budgeting, Board Governance, Compliance, Reporting	LC management; depreciation; AGM/resolution register; full analytics dashboard

Each phase ends with: full test suite pass (≥80% coverage), user acceptance testing on staging, production deployment, and user training for that phase's modules.

14. Key Technical Decisions

Decision	Chosen	Rejected	Rationale
Architecture	Modular Monolith	Microservices	Cross-module ACID transactions; 50–200 users; small team; lower ops overhead
HTTP Adapter	Express	Fastify	Larger middleware ecosystem; simpler file upload; Pino logger added manually
ORM	Prisma v5	TypeORM / raw SQL	Multi-schema support; type-safe queries; migrations as init container
State Management	TanStack Query + Zustand	Redux Toolkit	Clear server vs client state separation; less boilerplate; better caching
Job Queue	BullMQ + Redis	RabbitMQ / Kafka	Same process; sufficient at this scale; simpler infra; Bull Board UI
PDF Generation	Puppeteer (headless Chrome)	PDFKit / WeasyPrint	Pixel-perfect HTML rendering; supports Bangla fonts natively
Authentication	Argon2id + JWT + httpOnly	Sessions / bcrypt	Stateless for K8s; httpOnly cookie for refresh security; industry standard
File Storage	S3 / MinIO	Database BLOB / disk	Scalable; cost-effective; lifecycle policies; CDN-ready; never on local disk
Internationalisation	i18next (lazy-loaded)	Hardcoded strings	Bangladesh market requires Bangla; lazy loading keeps bundle size small
SMS Provider	SSL Wireless	Twilio / Infobip	Best OTP delivery rates in Bangladesh; local gateway; lower cost per SMS

15. Appendix — Design Document Inventory

The following documents form the complete design package for the OK Footwear ERP system. All files are provided alongside this document.

Document	File	Contents
Feature List	OK_Footwear_ERP_Feature_List.md	15 modules, 200+ features, phased delivery plan
Business Requirements (BRD)	OK_Footwear_ERP_BRD.md	Stakeholders, functional and non-functional requirements
Software Requirements (SRS)	OK_Footwear_ERP_SRS.md	IEEE 830 SRS with 14 functional requirement groups
Database Schema DDL	OK_Footwear_ERP_Schema.sql	Complete PostgreSQL DDL — 67 tables, indexes, triggers, functions
Schema Reference	OK_Footwear_ERP_Schema_Reference.md	2,362 lines — every table and field documented with descriptions
System Architecture Diagram	OK_Footwear_ERP_System_Architecture.html	Interactive full-system diagram; hover any component for details
Backend Architecture Diagram	OK_Footwear_ERP_Backend_Architecture.html	Interactive backend diagram with module implementation details
Testing Strategy	OK_Footwear_ERP_Testing_Strategy.md	Full test pyramid, tools, CI/CD integration, coverage targets
Unit Test Cases	OK_Footwear_ERP_Unit_Tests.md	42 unit test cases across all 8 NestJS modules
Integration Test Cases	OK_Footwear_ERP_Integration_Tests.md	28 HTTP integration test cases with real PostgreSQL + Redis
Database Test Cases	OK_Footwear_ERP_Database_Tests.md	26 trigger, function, and constraint test cases
Frontend Test Cases	OK_Footwear_ERP_Frontend_Tests.md	27 component and hook test cases with MSW API mocking
E2E Test Cases	OK_Footwear_ERP_E2E_Tests.md	18 Playwright full-journey test cases covering all critical flows
Contract Test Cases	OK_Footwear_ERP_Contract_Tests.md	10 Pact consumer-driven API contract test cases
Performance Test Cases	OK_Footwear_ERP_Performance_Tests.md	9 k6 load, stress, and spike test scenarios with SLA thresholds
Security Test Cases	OK_Footwear_ERP_Security_Tests.md	22 authentication, RBAC, injection, and session security tests
NestJS Architecture	OK_Footwear_ERP_NestJS_Architecture.md	Folder structure, module layout, key file implementations, naming conventions
This Document	OK_Footwear_ERP_Technical_Design_Document.pdf	Complete technical design document for management review

— End of Document —